

José Ignacio Orlando, Ph.D.



CONTACT INFORMATION	Associate Researcher CONICET PLADEMA / UNICEN Gral. Pinto 399 7000 Tandil, Buenos Aires, Argentina	Work: +54 249 438 5690 (Int 2411) E-mail: jjorlando@pladema.exa.unicen.edu.ar WWW: ignaciorlando.github.io
SUMMARY	Researcher in artificial intelligence and computer vision in medicine. University educator in artificial intelligence and applied mathematics. More than 10 years of academic and professional experience developing AI-based solutions for real problems in medicine, with more emphasis in ophthalmology. Passionate about coordinating working groups and educating new independent talent on solving challenging problems with innovative machine learning tools.	
CURRENT AFFILIATIONS	Associate Researcher, CONICET Since November 2019 • PLADEMA Institute, Universidad Nacional del Centro de la Provincia de Buenos Aires, Tandil, Argentina. • Promoted to Associate from Assistant in October 2023. Head of Practical Works, UNICEN Since December 2022Fundamentos de Ciencias de Datos, Taller de Matemática Computacional, Departamento de Computación y Sistemas, Facultad de Ciencias Exactas, Universidad Nacional del Centro de la Provincia de Buenos Aires (UNICEN), Tandil, Argentina. Profesor, UNICEN Desde Junio 2025Aprendizaje de máquina, Diplomatura Universitaria en Inteligencia Artificial (DUIA). Visión Computacional Basada en Inteligencia Artificial, Diplomatura Universitaria en Inteligencia Artificial (DUIA). Invited Professor, UBA Since April 2022Visión Computacional Basada en Redes Neuronales, Maestría en Explotación de Datos y Descubrimiento del Conocimiento, Facultad de Ciencias Exactas y Naturales, Universidad Nacional de Buenos Aires (UBA), CABA, Argentina.	
PREVIOUS AFFILIATIONS	Postdoctoral Research Associate, OPTIMA January 2018 - September 2019 • Christian Doppler Laboratory for Ophthalmic Image Analysis (OPTIMA), Department of Ophthalmology and Optometry, Medical University of Vienna, Vienna, Austria. Postdoctoral Research Associate, CONICET September 2017 to January 2018 • Pladema Institute, Facultad de Ciencias Exactas, UNICEN, Argentina. • CCT Tandil, CONICET, Argentina. Teaching Assistant, UNICEN 2010-2018; 2020-2022Facultad de Ciencias Exactas Ph.D Student, CONICET April 2013 to September 2017 • Pladema Institute, Facultad de Ciencias Exactas, UNICEN, Argentina. • CCT Tandil, CONICET, Argentina.	

	Intern, Inria Saclay - Team GALEN May-November 2013 Center for Learning and Visual Computing (CVN), Centrale Supélec, Paris, France.
EDUCATION	Universidad Nacional del Centro de la Provincia de Buenos Aires (UNICEN), Tandil, PBA, Argentina Ph.D., Computational and Industrial Mathematics, April 2013 - September 2017, <i>summa cum laude</i> - Thesis: <i>Machine learning for ophthalmic screening and diagnostics from fundus images</i> - Advisors: Dr. Matthew B. Blaschko and Dr. Mariana del Fresno - Area of Study: Medical imaging M.S., Software Engineering, March 2013 - Thesis Topic: <i>Texture based segmentation of 3D medical images</i> - Advisor: Dr. Mariana del Fresno - Area of Study: Image and Signal Processing
TEACHING	Universidad Nacional del Centro de la Provincia de Buenos Aires, Tandil, PBA; Argentina <i>Professor</i> 2020 Machine Learning (Aprendizaje de máquina) (2020-Today). Diplomatura Universitaria en Inteligencia Artificial. Facultad de Ciencias Exactas. Computer Vision with Artificial Intelligence (Visión Computacional Basada en Inteligencia Artificial) (2020-Today). Diplomatura Universitaria en Inteligencia Artificial. Facultad de Ciencias Exactas. <i>Teaching Assistant</i> March 2010-2022 Workshop on Computational Mathematics (2021-2024) Information Theory (2020) Workshop on Computational Mathematics (2015-2017) Information Theory (2013-2016) Workshop on Medical Imaging (2014-2015) Software Development Methodologies (2010-2012) Escuela de Educación Técnica N. 1, Las Flores, PBA; Argentina <i>Professor</i> 2010-2013 Operating Systems Laboratory Artificial Intelligence Network Systems Implementation
METRICS	Google Scholar: H-index = 25. i10 index = 27. Total number of citations: 4073. (Last checked: July 14, 2026)
REFEREED JOURNAL PUBLICATIONS	[1] A. Luisi, F. Orte, E. Wolfram, and J. I. Orlando. Aplicación de un modelo de machine learning para clasificar la nubosidad a partir de datos de irradiancia global horizontal. <i>Avances en Energías Renovables y Medio Ambiente - AVERMA</i>, 29:524–532, 2026. https://portalderevistas.unsa.edu.ar/index.php/averma/article/view/5286.

- [2] L. G. Telesco, D. Nejamkin, E. Mata, F. Filizzola, K. Wignall, L. Franco Troilo, M. de los Ángeles Cenoz, M. Thompson, M. Leguía, I. Larrabide, and *J. I. Orlando*. Semi-supervised multi-task learning for interpretable quality assessment of fundus images. *Biomedical Signal Processing and Control*, 113, 109167 (2026). doi:10.1016/j.bspc.2025.109167.
- [3] S. Vitale, *J. I. Orlando*, E. Iarussi, A. Díaz, I. Larrabide. Improving Realism in Abdominal Ultrasound Simulation Combining a Segmentation-Guided Loss and Polar Coordinates training. *Medical Physics*. In press. (2025) doi:10.1002/mp.17801. IF: 3.2 (2023). Scimago Quartile: Q1.
- [4] D. Braggio, H. C. Külsgaard, M. Vallejo-Azar, M. Bendersky, P. González, L. Alba-Ferrara, *J. I. Orlando*, Ignacio Larrabide. A Self-supervised Deep Learning Model for Diagonal Sulcus Detection with Limited Labeled Data. *Neuroinformatics*, 23, 1–17 (2025). doi:10.1007/s12021-024-09700-7. IF: 2.7 (2024). Scimago Quartile: Q1.
- [5] A. R. Lusi, P. F. Orte, E. Wolfram, *J. I. Orlando*. Cloud classification through machine learning and global horizontal irradiance data analysis. *Quarterly Journal of the Royal Meteorological Society*, 150(765). 2024. doi:10.1002/qj.4880. IF: 16.8 (2023). Scimago Quartile: Q1.
- [6] N. Dazeo, *J. I. Orlando*, C. García, R. Muñoz, L. Obrado, H. Fernández, J. Blasco, L. San Román, J. M. Macho, A. Ding, R. Utz, and I. Larrabide. Computer Aided Intracranial Aneurysm Treatment Based on 2D/3D Mapping, Virtual Deployment and Online Distal Marker Detection. *Cardiovascular Engineering and Technology*, In press. 2024. doi:10.1007/s13239-024-00745-y. IF: 1.6 (2023). Scimago Quartile: Q3.
- [7] P. Seeböck, *J. I. Orlando*, M. Michl, J. Mai, U. Schmidt Erfurth, and H. Bogunović. Anomaly guided segmentation: Introducing semantic context for lesion segmentation in retinal OCT using weak context supervision from anomaly detection. *Medical Image Analysis*, 93(103104). 2024. doi:10.1016/j.media.2024.103104. IF: 13.828 (2024). Scimago Quartile: Q1.
- [8] H. Fang, F. Li, J. Wu, H. Fu, X. Sun, *J. I. Orlando*, ... and Y. Xu. (2024). Open fundus photograph dataset with pathologic myopia recognition and anatomical structure annotation. *Scientific Data*, 11(1), 99. 2024. doi:10.1038/s41597-024-02911-2. IF: 9.8 (2024). Scimago Quartile: Q1.
- [9] J. Wu, H. Fang, F. Li, H. Fu, F. Lin, J. Li, ... *J. I. Orlando*, X. Zhang, and Y. Xu. GAMMA Challenge: Glaucoma grAding from Multi-Modality imAges. *Medical Image Analysis*, 90(102938). 2023. doi:10.1016/j.media.2023.102938. IF: 10.9 (2023). Scimago Quartile: Q1
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- [11] L. Rocamora, *J. I. Orlando*, C. Lwowski, T. Kohnen, E. Mertens and K. van Keer. Postoperative vault prediction for phakic implantable collamer lens surgery: LASSO formulas. *Journal of Cataract & Refractive Surgery*, 49 (2), 126. 2022. doi:10.1097/j.jcrs.0000000000001079. IF: 3.351 (2022). Scimago Quartile: Q1.

- [12] H. Fang, F. Li, H. Fu, X. Sun, X. Cao, F. Lin, J. Son, S. Kim, G. Quellec, S. Matata, S. M. Shankaranarayana, Y-T. Chen, C-h. Wang, N. A. Shah, C-Y. Lee, C-C. Hsu, H. Xie, B. Lei, U. Baid, S. Innani, K. Dang, W. Shi, R. Kamble, N. Singhal, C-W. Wang, S-C. Lo, *J. I. Orlando*, H. Bogunović, X. Zhang and Y. Xu. ADAM Challenge: Detecting Age-related Macular Degeneration from Fundus Images. *IEEE Transaction on Medical Imaging*, In press, 2022.
doi:[10.1109/TMI.2022.3172773](https://doi.org/10.1109/TMI.2022.3172773). IF: 13.94 (2020). Scimago Quartile: Q1.
- [13] D. Hofer, U. Schmidt-Erfurth, *J. I. Orlando*, F. Goldbach, B. S. Gerendas and P. Seeböck. Improving Foveal Avascular Zone Segmentation in Fluorescein Angiograms by Leveraging Manual Vessel Labels from Public Color Fundus Pictures. *Biomedical Optics Express*, 13(5): 2566-2580, 2022.
doi:[10.1364/BOE.452873](https://doi.org/10.1364/BOE.452873). IF: 3.732 (2022). Scimago Quartile: Q1.
- [14] P. Seeböck, W-D Vogl, S. M. Waldstein, *J. I. Orlando*, M. Baratsits, T. Alten, M. Arikian, G. Mylonas, H. Bogunović and U. Schmidt-Erfurth. Linking Function and Structure with ReSenseNet: Predicting Retinal Sensitivity from Optical Coherence Tomography using Deep Learning. *Ophthalmology Retina*, In press, 2022.
doi:[10.1016/j.oret.2022.01.021](https://doi.org/10.1016/j.oret.2022.01.021). IF: 3.9 (2022). Scimago Quartile: Q1.
- [15] P. Navarro, *J. I. Orlando*, C. Delrieux and E. Iarussi. SketchZooms: Deep Multi-view Descriptors for Matching Line Drawings. *Computer Graphics Forum*, 40(1): 410-423, 2020.
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- [17] H. Külsgaard, *J. I. Orlando*, M. Bendersky, J. P. Princich, L. S. R. Manzanera, A. Vargas, S. Kochen and I. Larrabide. Machine learning for filtering out false positive grey matter atrophies in single subject voxel based morphometry: a simulation based study. *Journal of the Neurological Sciences*, 420: 117220, 2020.
doi:[10.1016/j.jns.2020.117220](https://doi.org/10.1016/j.jns.2020.117220). IF: 3.115 (2019). Scimago Quartile: Q2.
- [18] H. Fu, F. Li, X. Sun, X. Cao, J.Liao, *J. I. Orlando*, ..., Y. Xu. AGE Challenge: Angle Closure Glaucoma Evaluation in Anterior Segment Optical Coherence Tomography. *Medical Image Analysis*, 66: 101798, 2020.
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- [19] *J. I. Orlando*, B.S. Gerendas, S. Riedl, C. Grechenig, A. Breger, M. Ehler, S.M. Waldstein, H. Bogunović and U. Schmidt-Erfurth. Automated Quantification of Photoreceptor alteration in macular disease using Optical Coherence Tomography and Deep Learning. *Scientific Reports*. 10, 2020.
doi:[10.1038/s41598-020-62329-9](https://doi.org/10.1038/s41598-020-62329-9). IF: 4.12 (2018). Scimago Quartile: Q1.
- [20] *J. I. Orlando*, H. Fu, J. Barbossa Breda, K. van Keer, D. R. Bathula, A. Díaz Pinto, R. Fang, ..., H. Bogunović. REFUGE Challenge: A Unified Framework for Evaluating Automated Methods for Glaucoma Assessment from Fundus Photographs. *Medical Image Analysis*, 59, 2020.
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- [21] P. Seeböck, *J.I. Orlando*, T. Schlegl, S.M. Waldstein, H. Bogunović, S. Klimescha, G. Langs and U. Schmidt-Erfurth. Exploiting Epistemic Uncertainty of Anatomy Segmentation for Anomaly Detection in Retinal OCT. *IEEE Transactions in Medical Imaging*, 39(1):87–98, 2020.
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- [22] D. Romo-Bucheli, P. Seeböck, *J. I. Orlando*, B.S. Gerendas, S.M. Waldstein, U. Schmidt-Erfurth, H. Bogunović. Reducing image variability across OCT devices with unsupervised unpaired learning for improved segmentation of retina. *Biomedical Optics Express*, 11(1):346–363, 2020.
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- [23] A. Breger, *J. I. Orlando*, P. Harar, M. Dörfler, S. Klimescha, C. Grechenig, B. S. Gerendas, U. Schmidt-Erfurth, M. Ehler. On Orthogonal Projections for Dimension Reduction and Applications in Augmented Target Loss Functions for Learning Problems. *Journal of Mathematical Imaging and Vision*, 62, 376–394, 2020.
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- [24] S. Vitale, *J.I. Orlando*, E. Iarussi and I. Larrabide. Improving realism in patient-specific abdominal Ultrasound simulation using CycleGANs. *International Journal of Computer Assisted Radiology and Surgery*, 15, 183–192, 2020.
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- [26] *J.I. Orlando*, K. van Keer, J. Barbosa-Breda, H.L. Manterola, M.B. Blaschko, and A. Clausse. Proliferative Diabetic Retinopathy Characterization based on Fractal Features: Evaluation on a Publicly Available Data Set. *Medical Physics*, 44(12):6425–6434, 2017.
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- [27] *J.I. Orlando*, E. Prokofyeva, and M.B. Blaschko. A discriminatively trained fully connected conditional random field model for blood vessel segmentation in fundus images. *IEEE Transaction on Biomedical Engineering*, 64(1):16–27, 2017.
doi:10.1109/TBME.2016.2535311. IF: 4.288 (2017). Scimago Quartile: Q1.
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doi:10.1007/s11548-015-1345-4. IF: 1.961 (2014). Scimago Quartile: Q2.
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Indexed: Latindex.
- [30] E. Moris and *J. I. Orlando*. Can Coding Agents Build Robust Baselines? A Skill-Based Approach for Automating the Medical Imaging Model-Development Pipeline. In: *2nd Agentic AI for Medicine Workshop at MICCAI 2026 (Agentic-Med)*, 2026.
<https://arxiv.org/abs/2608.23336>

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PUBLICATIONS

- [31] F. J. Arellano and J. I. Orlando. Evaluating Fundus-Specific Foundation Models for Diabetic Macular Edema Detection. In: *International Symposium on Medical Information Processing and Analysis (SIPAIM 2025)*, IEEE. 2025. In press.
- [32] E. Moris, I. Larrabide, and J.I. Orlando. Semi-supervised learning with Noisy Students improves domain generalization in optic disc and cup segmentation in uncropped fundus images. In: *Medical Imaging with Deep Learning (MIDL 2024)*, PMLR. 2024. In press.
- [33] D. Deangeli, E. Iarussi, J.P. Princich, M. Bendersky, I. Larrabide and J.I. Orlando. Learning normal asymmetry representations for homologous brain structures. In: *26th International Conference in Medical Imaging Computing and Computer-Assisted Intervention (MICCAI 2023)*, LNCS. p. 77-87. 2023. doi:[10.1007/978-3-031-43993-3_8](https://doi.org/10.1007/978-3-031-43993-3_8)
- [34] T. Castilla, M.S. Martínez, M. Leguía, I. Larrabide and J.I. Orlando. A ResNet is All You Need? Modeling A Strong Baseline for Detecting Referable Diabetic Retinopathy in Fundus Images. In: *18th International Symposium on Medical Information Processing and Analysis (SIPAIM 2022)*, vol. 12567, pp. 212-221. 2022. doi:[10.1117/12.2669816](https://doi.org/10.1117/12.2669816)
- [35] E. Moris, N. Dazeo, M.P. Albina de Rueda, F. Filizzola, N. Iannuzzo, D. Nejamkin, K. Wignall, M. Leguía, I. Larrabide and J.I. Orlando. Assessing Coarse-to-Fine Deep Learning Models for Optic Disc and Cup Segmentation in Fundus Images. In: *18th International Symposium on Medical Information Processing and Analysis (SIPAIM 2022)*, vol. 12567, pp. 232-241. 2022. doi:[10.1117/12.2670093](https://doi.org/10.1117/12.2670093)
- [36] C. García, Y. Fang, J. Liu, A.P. Narata, J.I. Orlando and I. Larrabide. A deep learning model for brain vessel segmentation in 3DRA with arteriovenous malformations. In: *18th International Symposium on Medical Information Processing and Analysis (SIPAIM 2022)*, vol. 12567, 61-69. 2022. doi:[10.1117/12.2669916](https://doi.org/10.1117/12.2669916)
- [37] J.I. Orlando, A. Breger, H. Bogunović, S. Riedl, B.S. Gerendas, M. Ehler and U. Schmidt-Erfurth. An amplified-target loss approach for photoreceptor layer segmentation in pathological OCT scans. En: *6th MICCAI International Workshop on Ophthalmic Medical Image Analysis (OMIA 2019)*, Lecture Notes in Computer Science, vol 11855, 26-34. 2019. doi:[10.1007/978-3-030-32956-3_4](https://doi.org/10.1007/978-3-030-32956-3_4)
- [38] D. Hofer, J.I. Orlando, P. Seeböck, G. Mylonas, F. Goldbach, A. Sadeghipour, B.S. Gerendas and U. Schmidt-Erfurth. Foveal avascular zone segmentation in clinical routine fluorescein angiographies using multitask learning. En: *6th MICCAI International Workshop on Ophthalmic Medical Image Analysis (OMIA 2019)*, Lecture Notes in Computer Science, vol 11855, 35-42. 2019. doi:[10.1007/978-3-030-32956-3_5](https://doi.org/10.1007/978-3-030-32956-3_5)
- [39] R. Asgari, J.I. Orlando, S. Waldstein, F. Schlanitz, M. Baratsits, U. Schmidt-Erfurth and H. Bogunović. Multiclass segmentation as multitask learning for drusen segmentation in retinal optical coherence tomography. En: *Medical Image Computing and Computer Assisted Intervention (MICCAI 2019)*, Lecture Notes in Computer Science, vol. 11764, 192-200, 2019. doi:[10.1007/978-3-030-32239-7_22](https://doi.org/10.1007/978-3-030-32239-7_22)

- [40] *J.I. Orlando, P. Seeböck, H. Bogunovic, S. Klimesch, C. Grechenig, S. Waldstein, B.S. Gerendas and U. Schmidt-Erfurth. U2-Net: A Bayesian U-Net model with epistemic uncertainty feedback for photoreceptor layer segmentation in pathological OCT scans. In: 2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019), 1441-1445, 2019. doi:10.1109/ISBI.2019.8759581*
- [41] *P. Seeböck, D. Romo Bucheli, S. Waldstein, H. Bogunovic, J.I. Orlando, B.S. Gerendas, G. Langs and U. Schmidt-Erfurth. Using CycleGANs for effectively reducing image variability across OCT devices and improving retinal fluid segmentation. In: 2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019), 605-609, 2019. doi:10.1109/ISBI.2019.8759158*
- [42] *J.I. Orlando, J. Barbosa-Breda, K. van Keer, M.B. Blaschko, P.J. Blanco and C.A. Bulant. Towards a glaucoma risk index based on simulated hemodynamics from fundus images. In: Medical Image Computing and Computer Assisted Intervention (MICCAI 2018), Lecture Notes in Computer Science, 11071, 65-73, 2018. doi:10.1007/978-3-030-00934-2_8*
- [43] *J.I. Orlando, M. Fracchia, V. del Río, and M. del Fresno. Retinal blood vessel segmentation in high resolution fundus photographs using automated feature parameter estimation. In: 13th International Symposium on Medical Information Processing and Analysis (SIPAIM 2017), Proc. SPIE, 10572, 1057210, 2017. doi:10.1117/12.2283539*
- [44] *J.I. Orlando, E. Prokofyeva, M. del Fresno, and M.B. Blaschko. Convolutional neural network transfer for automated glaucoma identification. In: 12th International Symposium on Medical Information Processing and Analysis (SIPAIM 2016), Proc. SPIE, 10160, 101600U-101600U-10, 2017. doi:10.1117/12.2255740*
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- [46] *J.I. Orlando, H.L. Manterola, E. Ferrante, and F. Ariel. Arabidopsis roots segmentation based on morphological operations and CRFs. arXiv:1704.07793. Presented at ECImag 2014 (School and Workshop on Image Sciences), August 11–15, 2013.*

ABSTRACTS, POSTERS AND PRESENTATIONS IN INTERNATIONAL CONFERENCES

- [47] *R. Asgari, J.I. Orlando, S. Waldstein, F. Schlanitz, M. Baratsits, U. Schmidt-Erfurth and H. Bogunović. Multiclass segmentation as multitask learning for drusen segmentation in retinal optical coherence tomography. In: Medical Image Cluster (MIC) Festival 2019, 2019, Vienna, Austria. Poster presentation. Presenter: R. Asgari.*
- [48] *D. Hofer, J.I. Orlando, P. Seeböck, B. Gerendas and U. Schmidt-Erfurth. Foveal avascular zone segmentation in clinical routine fluorescein angiographies using multitask learning. In: Medical Image Cluster (MIC) Festival 2019, 2019, Vienna, Austria. Poster presentation. Presenter: D. Hofer.*
- [49] *S. Vitale, J.I. Orlando, E. Iarussi, and I. Larrabide. Improving realism in patient-specific abdominal Ultrasound simulation using CycleGANs. In: CARS 2019, 2019, Rennes, France. Poster presentation. Presenter: S. Vitale.*

- [50] P. Seeböck, W.-D. Vogl, S.M. Waldstein, M. Baratsits, *J.I. Orlando*, T. Alten, H. Bogunović, M. Arikan, G. Mylonas and U. Schmidt-Erfurth. Linking Function and Structure: Prediction of Retinal Sensitivity in AMD from OCT using Deep Learning. In: *ARVO 2019 Annual Meeting*, 2019. Poster presentation. Presenter: Phillip Seeböck (PhD Student, OPTIMA, Medical University of Vienna).
- [51] A. Breger, M. Ehler., B.S. Gerendas, *J.I. Orlando*, U. Schmidt-Erfurth. Dimension reduction in learning tasks. In: *90th GAMM Annual Meeting*, 2019. Poster presentation. Presenter: Anna Breger (PhD Student, Department of Mathematics, University of Vienna).
- [52] A. Breger, *J.I. Orlando*, M. Ehler. Learning and Dimension Reduction in Medical Image Analysis. In: *SIAM Conference on Imaging Sciences*, 2018. Poster presentation. Presenter: Anna Breger (PhD Student, Department of Mathematics, University of Vienna).

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- [54] *J.I. Orlando*, L.G. Telesco, F. Arellano, A. Orbe, M. Iparraguirre, A. Díaz, M. Leguía and I. Larrabide. Desempeño diagnóstico de una herramienta argentina de inteligencia artificial para detectar retinopatía diabética referible en programas de tamizaje. In: *XXV Congreso Argentino de Diabetes*, Hotel Provincial, Mar del Plata, Argentina, October 29–31, 2026. Accepted e-poster.
- [55] *J.I. Orlando*, M. Leguía, I. Larrabide. Retinar: Inteligencia Artificial y Telemedicina Distribuida para la Detección Temprana de la Retinopatía Diabética”. In: *IADI 2024, 1 Jornada Nacional de Inteligencia Artificial en Diagnóstico por Imágenes*, 2024. Oral presentation.
- [56] D. Deangeli, F. Iarussi, H. Külsgaard, D. Braggio, J.P. Princich, M. Bendersky, E. Iarussi, I. Larrabide and *J.I. Orlando*. Un índice basado en inteligencia artificial para la cuantificación de desviaciones de la asimetría normal del hipocampo a partir de resonancia magnética cerebral. In: *CADI 2022, Congreso Argentino de Diagnóstico por Imagen*, 2022. Poster.
- [57] S. Vitale, *J.I. Orlando*, E. Iarussi, A. Díaz and I. Larrabide. SIMECO: Un simulador de ecografías realistas basado en algoritmos de trazado de rayos e inteligencia artificial. In: *CADI 2022, Congreso Argentino de Diagnóstico por Imagen*, 2022. Poster.
- [58] C. García, A.P. Narata, Y. Fang, *J.I. Orlando* and I. Larrabide. Extracción automática de la vasculatura cerebral en imágenes de ATC. In: *CADI 2022, Congreso Argentino de Diagnóstico por Imagen*, 2022. Poster.
- [59] E. Moris, I. Larrabide and *J.I. Orlando*. Estimación automática de la relación vertical copa-disco mediante redes neuronales de segmentación a partir de fotografías de fondo de ojo. In: *CADI 2022, Congreso Argentino de Diagnóstico por Imagen*, 2022. Digital Poster.
- [60] *J.I. Orlando*, T. Castilla, A. Koch, I. Larrabide, M. Martínez y M. Leguía. Evaluación retrospectiva de un primer modelo de inteligencia artificial argentino para tamizaje automático de retinopatía diabética referible a partir de fotografías de

fondo de ojo. In: *CNO22, Congreso Nacional de Oftalmología 2022*, 2022. Oral presentation.

- [61] D. Braggio, H. Külsgaard, *J.I. Orlando*, J.P. Princich, M. Bendersky, S. Kochen and I. Larrabide. Caracterización morfológica de poblaciones cerebrales de control mediante autocodificadores con convoluciones basadas en grafos. In: *IA@Litoral: 1ras Jornadas de Inteligencia Artificial del Litoral*, 2019. Poster.
- [62] S. Vitale, *J.I. Orlando*, E. Iarussi and I. Larrabide. Mejorando el realismo de un simulador de ultrasonido abdominal paciente-específico utilizando CycleGANs. In: *IA@Litoral: 1ras Jornadas de Inteligencia Artificial del Litoral*, 2019. Poster.
- [63] *J.I. Orlando*, H.L. Manterola, E. Ferrante, and M. del Fresno. Un esquema de superficies activas basado en texturas para la segmentación de tumores cerebrales en MRI. In: *Proceedings of the 1st National Conference on Software Engineering / Information Systems (CONAIIISI 2013)*, 2013. Full article (in Spanish), oral presentation.
- [64] *J.I. Orlando*, and M. del Fresno. Segmentación automática de tejidos cerebrales en MRI multiespectrales mediante clasificación por mínima distancia euclídea. In: *Proceedings of the Argentine Conference on Informatics and Health (CAIS 2013)*, 2013. Full article (in Spanish), oral presentation.
- [65] H.L. Manterola, *J.I. Orlando*, E. Ferrante, and M. del Fresno. Extracción de objetos en imágenes médicas tridimensionales basada en características texturales. In: *Proceedings of the Argentine Conference on Informatics and Health (CAIS 2013)*, 2013. Full article (in Spanish), oral presentation.
- [66] *J.I. Orlando*, H.L. Manterola, E. Ferrante, and M. del Fresno. Detección de tumores en MRI basada en características texturales. In: *Proceedings of the School and Workshop on Image Sciences (ECImag 2012)*, 2012. Full article (in Spanish), oral presentation.
- [67] *J.I. Orlando*, H.L. Manterola, E. Ferrante, and M. del Fresno. Un enfoque híbrido para la segmentación de tumores en MRI cerebrales. In: *Proceedings of the Argentine Conference on Informatics and Health (CAIS 2012)*, 2012. Full article (in Spanish), oral presentation.

CONSULTING

CONICET ST6329 - Hired by Arionkoder Global LLC,
Director of AI Labs.

2023-ongoing

- Participation in pre-sales activities with clients interested in AI adoption.
- Design of AI-based solutions for industry problems.
- Effort estimation and identification of required human resources for project execution.
- Conducting technical interviews for ML Engineer and Data Scientist positions.
- Creation of articles and marketing materials, including one-pagers, presentations, and a podcast with four episodes.
- Jury member in hackathons and participation as an SME in AI/ML for innovation-related activities.
- Participation as an SME in AI/ML during Design Sprints with clients.
- Planning and coordination of internal initiatives (R&D area, paper reading group).
- General coordination of the execution of multiple client projects.

CONICET ST6329 - Hired by Lamansys SAS,

2023

Consultancy for developing a POC for the Ministry of Education of Argentina for detecting school dropout using AI.

- Design of the proposed solution.

- Coordination of activities with a doctoral student responsible for the solution's implementation.
- Joint data analysis.
- Report on the developed solution, shortcomings, and a roadmap for data collection to carry it out.

CONICET STAN 5948 - Hired by Arionkoder Global LLC, 2022-2023
Subject Matter Expert in ML/AI, to assist their AI processes.

- Creation of AI/ML content for the company's blog for positioning and digital marketing.
- Conducting technical interviews for ML Engineer and Data Scientist positions.

CONICET STAN 5887 - Hired by UNICEN, 2021-2023
Advising in AI and medical image processing and analysis for computer-assisted treatment with STENTs.

- Design of a deep learning-based solution for detecting radiopaque markers in 2D X-ray images.
- Design of the data annotation protocol for generating labels for algorithm training.
- General coordination of project activities, in collaboration with a doctoral student responsible for development and a researcher as an SME.

TECH TRANSFER

retinar. AI-based platform for diabetic retinopathy detection. 2019-ongoing

- Lead by myself, this project seeks to develop a telemedicine platform for diabetic retinopathy screening based on AI models.

SIMECO. Abdominal Ultrasound Simulator based on ray-casting techniques and artificial intelligence. 2019-ongoing

- Lead by Santiago Vitale, this project seeks to develop a cutting-edge realistic abdominal ultrasound simulator for training clinicians in disease diagnosis and treatment.

ADVISING AND MENTORING

PhD students

- **Franco Arellano**, 2024, PhD in Computational and Industrial Mathematics (UNICEN)
Topic: Foundation Models in Retinal Image Analysis.
Advisor: *José Ignacio Orlando*.
- **Lucas Telesco**, 2022, PhD in Computational and Industrial Mathematics (UNICEN)
Topic: Deep learning tools for computer-assisted diagnosis of diabetic retinopathy from color fundus photographs .
Advisor: *José Ignacio Orlando*.
- **Leandro Rocamora**, 2021, PhD in Computational and Industrial Mathematics (UNICEN)
Topic: Data science and artificial intelligence contributions to real problems in ophthalmology and meteorology.
Advisor: *José Ignacio Orlando*.
- **Duilio Deangeli**, 2021, PhD in Computational and Industrial Mathematics (UNICEN)
Thesis: Neuroimaging with deep learning: understanding morphological and functional changes in brain MRI due to aging and neurodegenerative conditions. Defended. Score: 10/10.
Advisor: Ignacio Larrabide Co-advisor: *José Ignacio Orlando*.

- **Eugenia Moris**, 2020–2024, PhD in Computational and Industrial Mathematics (UNICEN)
Thesis: Experimental modifications and new machine learning approaches for applications in sleep analysis and glaucoma detection. Defended in 2024. Score: 10/10.
Advisor: Ignacio Larrabide Co-advisor: *José Ignacio Orlando* (joined in 2021).
- **Santiago Vitale**, 2018–2023, PhD in Computational and Industrial Mathematics (UNICEN)
Thesis: Unpaired generative adversarial learning model for simulating realistic ultrasound abdominal images. Defended in 2023. Score: 10/10.
Advisor: Ignacio Larrabide Co-advisor: *José Ignacio Orlando* (joined in 2019).

Master students

- **Emmanuel Pengue**, 2026–2027, M.Sc. in Data Mining and Knowledge Discovery, Faculty of Exact and Natural Sciences and Faculty of Engineering, University of Buenos Aires (UBA)
Thesis: Integration of acoustic shadow masks into generative ultrasound models.
Advisor: *José Ignacio Orlando*.
Co-advisor: Santiago Vitale.
In progress.

Training in Research Scholarships

- **Franco Arellano**, 2023–2024, Beca de Entrenamiento a la Investigación BENTRE23 (Comisión de Investigaciones Científicas de la Provincia de Buenos Aires)
Plan de trabajo: smartScraper: Scraping images through web interfaces and anomaly detection for training deep learning models for ophthalmological applications.
Director: *José Ignacio Orlando*
- **Tomás Castilla**, 2021–2022, Beca INI de Ingreso a la Investigación (Programa de Fortalecimiento a la Ciencia y la Tecnología en las Universidades Nacionales)
Topic: Deep learning algorithms for computer assisted diagnostic of diabetic retinopathy from fundus photographs.
Advisor: José Ignacio Orlando
- **Francisco Iarussi**, 2020–2021, Beca EVC-CIN de Estímulo a las Vocaciones Científicas 2019
Topic: DeepBrain: Machine Learning applied to study brain morphological alterations in MRI scans.
Advisor: Ignacio Larrabide
Co-advisor: José Ignacio Orlando

Undergraduate Theses

- **Lautaro Barrassa**, 2026–2027, Software Engineering final project, Faculty of Exact Sciences, UNICEN (advisor)
Thesis topic: Deep learning for recognition and grading of diabetic retinopathy in ultra-widefield fundus images.
Co-advisor: Franco Arellano.
In progress.
- **Joaquín Ciappa**, 2026–2027, Software Engineering final project, Faculty of Exact Sciences, UNICEN (advisor)
Thesis topic: Solar irradiance prediction from meteorological data and cloud images using machine learning.
Co-advisor: Facundo Orte.
In progress.

- **Joaquín González Tesoro**, 2026–2027, Software Engineering final project, Faculty of Exact Sciences, UNICEN (advisor)
Thesis topic: Implementation of a web application to support people with diabetes.
Co-advisor: Ignacio Larrabide.
In progress.
- **Guadalupe Rapela**, 2026–2027, Software Engineering final project, Faculty of Exact Sciences, UNICEN (advisor)
Thesis topic: Implementation of harnessing strategies for agentification of the software development life cycle.
In progress.
- **Florencia Paglione**, 2026–2027, Software Engineering final project, Faculty of Exact Sciences, UNICEN (advisor)
Thesis topic: Development of a data recording platform for the wine industry.
Co-advisor: Rosana Cepeda, M.Sc.
In progress.
- **Julián Torrisi and Octavio Ituarte**, 2024–2025, Software Engineering final project (advisor)
Thesis topic: Full-stack resolution of technical debt, incorporation of new features, and redefinition of internal development processes for the Retinar platform
Co-advisor: Lucas Telesco.
Not defended yet.
- **Lautaro Gramuglia**, 2020-2021, Software Engineering thesis (director)
Thesis topic: Artery/Vein classification from color fundus photographs for blood flow simulations
Co-advisor: Carlos A. Bulant. Finished. Score: 10/10.
- **Tomás Castilla**, 2020–2021, Software Engineering thesis (director)
Thesis topic: Deep learning algorithms to support diabetic retinopathy screening on fundus photographs
Co-advisor: Ignacio Larrabide. Finished. Score: 10/10.
- **Rodrigo Cobo**, 2019–2020, Software Engineering thesis (director)
Thesis topic: Stereoscopic camera simulation using neural networks
Co-advisor: Ignacio Larrabide. Finished. Score: 10/10.
- **Francisco Iarussi**, 2020–2021, Software Engineering thesis (co-advisor)
Thesis topic: Characterization of hippocampal asymmetry using artificial intelligence techniques..
Advisors: Emmanuel Iarussi e Ignacio Larrabide
Finished. Score: 10/10.
- **Mauro Giamberardino, Ariel Borthiry**, 2016–2017, Software Engineering thesis (director)
Thesis topic: Retinal blood vessel segmentation in ultra-wide field of view angiographies
Finished. Score: 10/10.
- **Valeria del Río, Marcos Fracchia**, 2016–2017, Software Engineering thesis (director)
Thesis topic: Feature engineering for retinal blood vessel segmentation in fundus images
Finished. Score: 10/10.

- **Carmen Escudero Leoz, Manuel Corrales**, 2014, Software Engineering thesis (co-advisor)
Thesis director: Mariana del Fresno
Thesis topic: Integrating fuzzy c-means and deformable models for 3D medical image segmentation.
Finished. Score: 10/10.

GRANTS AND FUNDING AWARDED

PIP 2025-2027 - 11220250101273CO. "Intelligent agents based on language models and deep learning tools for automating diagnostic processes in ophthalmology"

Proyectos de Investigación Plurianuales de CONICET.

Project leader.

Grant: AR\$ 15.100.000.

FITBA 2024-276A. "Development of Retinar as an extension point of the Integrated Health Record for AI-based diabetic retinopathy screening"

- Fondo de Innovación Tecnológica de Buenos Aires (Ministerio de Producción, Ciencia y Tecnología, Provincia de Buenos Aires).

Project leader.

Budget awarded: AR\$ 32.000.000.

PICT 2021-00023. "retinar: artificial intelligence for computer-aided diagnostic of diabetic retinopathy"

- Proyectos de Investigación Científica y Tecnológica PICT 2021 Start-Up (FONCyT, Agencia I+D+i, Ministerio de Ciencia, Tecnología e Innovación).

Project leader.

Budget awarded: AR\$ 2.700.000.

PIP GI 2021-2023 - 11220200102472CO. "yatiris++: artificial intelligence for clinical applications based on medical images"

Proyectos de Investigación Plurianuales de CONICET para grupos de investigación.

Project leader

Grant: AR\$ 1.320.000.

NVIDIA Hardware Grant. "Self-supervised learning for phenotyping glaucoma from color fundus photographs"

Project leader

Grant: 400 hours on V100 GPU instances via SaturnCloud.

PN 376/2021. PAC Emprendedores para la Innovación. "retinar"

Non-refundable contribution granted by Secretaría de la Pequeña y Mediana Empresa y de los Emprendedores from Ministerio de Desarrollo Productivo (Argentina) to aid start-up companies to develop their products.

Project leader

Grant: AR\$ 1.173.000. Counterpart amount: AR\$ 207.000.

NVIDIA Applied Research Accelerator Program. "retinar: assisting remote diabetic retinopathy screening with AI tools"

NVIDIA program to collaborate in projects focused on technology transfer.

Project leader.

Award: 500 hours on V100 GPU instances via SaturnCloud.

PICT 2019-00070. "Characterization of optic nerve head morphology from color fundus pictures using deep learning"

Proyectos de Investigación Científica y Tecnológica PICT 2019 Joven Investigador (FONCyT, Agencia I+D+i, Ministerio de Ciencia, Tecnología e Innovación).

Project leader.

Budget awarded: \$ 475.000.

03-JOVIN-37C. "Towards a smart platform for remote diabetic retinopathy screening: quality control in fundus photographs using autoencoders"

Convocatoria Jóvenes Investigadores JOVIN 2020/2021 (Programa de Fortalecimiento a la Ciencia y la Tecnología en Universidades Nacionales, Secretaría de Ciencia, Arte y Tecnología, UNICEN).

Project leader.

Budget awarded: AR\$ 100.000.

Kaggle Open Data Research Grant 2020. "Weakly-supervised Abdominal Ultrasound Segmentation using Simulated Scans and Cycle-Consistency based Generative Models"

Kaggle Open Data Research Grant, donated by Google Kaggle.

Project leader.

Budget awarded: USD 2.000.

NVIDIA Hardware Grant. "Deep learning based segmentation of the photoreceptor layers in Spectral Domain Optical Coherence Tomography scans"

NVIDIA Hardware Grant, donated by NVIDIA Inc.

Project leader.

NVIDIA GTX Titan Xp (approx. price: €1.000).

MEMBERSHIP IN OTHER
RESEARCH PROJECTS

Estudio de factibilidad para el desarrollo de un sistema de diagnóstico a distancia y en tiempo real de muestras citológicas de extendidos cervicovaginales fijados (PAPs) asistido por Inteligencia Artificial, 2023-2025

CLIAS funding.

Research associate.

Project leader: Dr. Luciana Bruno.

Generative Networks for Interactive 2D/3D Design and Multiview Synthesis, 2020-2021

PID UTN SIUTNBA0005534.

Research associate.

Project leader: Dr. Emmanuel Iarussi.

CrossMatch: Detection of Cross-Domain Matchings in Sketches using Deep Learning, 2019-2019

PICT 2018-4517 (ANPCyT, Secretaría de Gobierno de Ciencia, Tecnología e Innovación Productiva, Ministerio de Educación, Ciencia y Cultura de la Nación, Argentina)

Research associate.

Project leader: Dr. Emmanuel Iarussi.

CrossMatch: Detection of Cross-Domain Matchings using Deep Learning, 2018-2020

Project SIUTNBA0005139 (Universidad Tecnológica Nacional - Regional CABA (UTN), Argentina)

Research associate.

Project leader: Dr. Emmanuel Iarussi.

HI-MED: Computational tools applied to image quantification, simulation and treatment planning in medicine, 2016-2019

PICT 2016-0116 (ANPCyT, Ministerio de Ciencia, Tecnología e Innovación Productiva, Argentina)

Research associate.

Project leader: Dr. Ignacio Larrabide.

Machine learning applied to detection and classification of emergency situations in remote sensing images, 2017-2018

Project A3SIS (Ministry of Defense, Argentina)
Research associate.
Project leader: Dr. Lucas Lo Vercio.

Development of models and applications for simulation, optimization, computer graphics and image processing, 2014-2018

Project 03/C259 (SPU, Ministry of Education, Argentina)
Research associate.
Supervisor: Dr. Mariana del Fresno.
Project leader: Dr. Marcelo Vénere.

Imaging and simulation models for vascular disease diagnosis and treatment, 2014-2016

PICT 2014-1730 (National Agency of Scientific and Technological Promotion, Ministry of Science, Argentina)
Research associate.
Supervisors: Dr. Mariana del Fresno and Dr. Matthew B. Blaschko.
Project leader: Dr. Ignacio Larrabide.

National strategy of articulation between the university and the secondary school for generating vocations and improving engineering and science education, 2014-2016

PCTI 121 (Ministry of Science, Argentina)
Collaborator.
Project leader: Prof. Mabel Pacheco.

3D digital image processing and segmentation for developing medical and industrial applications, 2013-2014

PICT 2010-1287 (National Agency of Scientific and Technological Promotion, Ministry of Science, Argentina)
Research associate.
Supervisor: Dr. Mariana del Fresno.
Project leader: Dr. Marcelo Vénere.

AWARDS

Winner, 16th Soluciones Innovadoras Banco Nación Contest (Argentina), 2025

- Award for Retinar as one of two winning innovative projects among more than 700 applications from Argentina.
- Organized by Fundación Empretec Argentina and Banco de la Nación Argentina. Prize awarded: AR\$ 12.000.000.

Special Award at Transformar Salud 2025-2026 (Argentina)

- Special recognition for Retinar as a project with continuity and growth after its previous participation, supporting implementation of AI-based ophthalmological screening in Tandil.
- Program organized by Fundación Garrahan and Roche Argentina. Economic support awarded: AR\$ 7.000.000.

Transformative AI - Award for AI innovation (Argentina), 2024

- Award for Retinar as the best scientific, technological and/or industrial AI innovation initiative of the country.
- Granted by Secretary of Innovation, Science and Technology, Argentina.

1st Prize at Transformar Salud 2023 (Argentina)

- Award for the best digital transformation proposal for healthcare with retinar, out teleophthalmology platform based on AI.

- Contest of projects and ideas for digital transformation in healthcare, organized by Fundación Garrahan and Roche Argentina.

First Prize at EconEmprende (Argentina)

- Prize for the best business project/idea at EconEmprende, organized by the Faculty of Economic Sciences at UNICEN, with our US simulator SIMECO. S. Vitale, C. Nagy, *J.I. Orlando*, and I. Larrabide.

Magna Cum Laude Prize at CADI 2022 (Argentina)

- Prize for the best poster at the Argentinian Conference of Diagnostic Imaging 2022 (CADI 2022), organized by the Argentine Society of Radiology, with the abstract "SIMECO: A realistic ultrasound simulator based on ray-tracing algorithms and artificial intelligence", S. Vitale, *J.I. Orlando*, E. Iarussi, A. Díaz and I. Larrabide.

Award of the National Academy of Medicine 2021. (Argentina)

- The National Academy of Medicine of Argentina awards the best original work made in Argentina on scientific and medical topics. In 2021 it was focused in Artificial Intelligence in Medicine. We were awarded for our work entitled "Development of a national artificial intelligence model for referable diabetic retinopathy screening from fundus photographs" ("Desarrollo de un modelo de inteligencia artificial nacional para el tamizado de casos de retinopatía diabética referible a partir de fotografías de fondo de ojo"), authored by J.I. Orlando, T. Castilla, A. Koch, M. Martínez, M. Leguía e I. Larrabide.

First Prize Oral Presentations at CNO22. (Argentina)

- First prize for the best oral presentation at the National Congress of Ophthalmology 2022 (CNO22), organized by the Argentine Council of Ophthalmology, with the abstract Retrospective evaluation of a first argentinian artificial intelligence model for automated screening of referable diabetic retinopathy from color fundus photographs", J.I. Orlando, T. Castilla, A. Koch, I. Larrabide, M. Martínez y M. Leguía.

Prendete 2020. Start-ups contest (<http://prendete.co/>). (Argentina)

- Member of retinar (<https://retinar.com.ar/en/>), one of the 15 projects selected as finalist among 130 initiative, and awarded for being the most voted by the public.

MICCAI Society

- MICCAI 2014 Student Travel Award.

Network of Software Engineering / Information Systems (RIISIC) (Argentina)

- Best paper award on students session, with the paper "Un enfoque de superficies activas basado en texturas para la segmentación de tumores en MRI cerebrales", J.I. Orlando, H.L. Manterola, E. Ferrante, and M. del Fresno.

Argentine Council of Engineering (CAI) (Argentina)

- Best engineering thesis, entitled "Segmentación de imágenes médicas tridimensionales basada en indicadores de texturas".

SCHOLARSHIPS AWARDED

- Ph.D. scholarship, Consejo Nacional de Investigaciones Científicas y Técnicas (National Council of Scientific and Technological Research), CONICET, Argentina. April 2013 to March 2018.
- Internship grant, Institut National de Recherche en Informatique et Automatique, INRIA, France. Équipe GALEN, INRIA Saclay, Palaiseau, France, and Center for Learning and Visual Computing, École Centrale Paris, France. Advisor: Dr. Matthew B. Blaschko. 6 months.

- Undergraduate research trainee scholarship, Comisión de Investigaciones Científicas de la Provincia de Buenos Aires, CIC-PBA, Argentina. Advisor: Dr. Mariana del Fresno. 4 months.

EVALUATION

Evaluation Committee CONICET

- 2021. External Specialist in evaluation for Convocatoria PIP 2021-2023 Grupo Investigación.
- 2020. Evaluator of Ingreso a la Carrera de Investigador Científico CONICET 2020 for the committee INGENIERÍA CIVIL, ELÉCTRICA, MECÁNICA E ING. RELACIONADAS PARA INGRESOS CIC.

Evaluation of research personnel

- 2021. Evaluator of 5 final reports of undergrad students who got a CIN Training-in-Research Scholarship in 2017. Secretaría de Ciencia, Arte y Tecnología, UNICEN.

Doctoral thesis

- 2024. Member of the examination committee of the PhD thesis proposal of José Morano García. Medical University of Vienna, Austria. Supervisor: Hrvoje Bogunovic.
- 2023. Member of the examination committee of the PhD thesis of Bioing. Ramiro Gatti. PhD in Engineering with mention in Bioengineering, Facultades de Ciencias de la Alimentación, Ciencias Agropecuarias e Ingeniería, UNER (Argentina). Supervisors: Dr. José Biurrun Manresa, Dr. Gabriel Gentiletti.
- 2023. Member of the examination committee of the MSc thesis of Mariana Vallejo-Azar. MSc. in Neurosciences. Universidad Nacional Arturo Jauretche (Argentina). Directores: Dr. Juan Pablo Princich, Dra. Paula González.
- 2022. Member of the examination committee of the PhD thesis of Lic. Adán Faramián. Doctorado en Ciencias Aplicadas Mención Ambiente y Salud. Facultad de Ciencias Exactas, UNICEN (Argentina). Supervisors: Dr. Raúl Rivas, Dr. Facundo Carmona.
- 2022. External academic evaluator of the PhD thesis of Roberto Romero. PhD in Information and Telecommunications Technologies, Universidad de Valladolid (España). Supervisors: Dr. María García Gadañón.
- 2022. Member of the examination committee of the PhD thesis of Bioeng. Agustina Larrazabal. Doctorado en Ingeniería, Mención Inteligencia Computacional, Señales y Sistemas. Facultad de Ingeniería y Ciencias Hídricas, Universidad Nacional del Litoral (Argentina). Supervisors: Dr. César Martínez, Dra. Cecilia García Cena.

Final projects and bachelor thesis

- 2020. Evaluator of thesis plan of M.S. Software Engineering, Facultad de Ciencias Exactas, UNICEN. Student: Valeria LUCHESSI. Supervisor: Dr. José Massa.

ACADEMIC SERVICE

Associate Editor

- Medical Image Analysis (IF: 10.9, Scimago Quartile: Q1), 2024.

Scientific committees

- Program Committee: 18th INTERNATIONAL SYMPOSIUM ON MEDICAL INFORMATION PROCESSING AND ANALYSIS (SIPAIM), Valparaíso, Chile, 9-11 Noviembre, 2022.
- Program Committee: 22do Congreso de Bioingeniería (SABI), Piriápolis, Uruguay, 4-6 March, 2020.
- Program Committee: LatinX in AI Research at NeurIPS 2019, Vancouver, Canada, 9 December, 2019.
- Organizer: Pathologic Myopia Challenge (PALM), at ISBI'19. Venice, Italy, 8 April, 2019.

- Scientific Committee: Workshop on GRaphs in biomedicAl Image anaLysis (GRAIL), at MICCAI'18. Granada, Spain, 20 September, 2018.
- Organizer: Retinal Fundus Glaucoma Challenge (REFUGE), at MICCAI'18. Granada, Spain, 20 September, 2018.
- Program Committee: Learning with Limited Data (LLD) workshop, at NIPS'17. Long Beach, CA, USA, 4th December, 2017.
- Scientific Committee: Workshop on GRaphs in biomedicAl Image anaLysis (GRAIL), at MICCAI'17. Quebec, Canada, 14 September, 2017.
- Local Organizing Committee: 12th Symposium on Medical Information Processing and Analysis (SIPAIM'16), Tandil, Argentina, 5-7 December, 2016.

Referee Service

- MICCAI 2018, 2019, 2020, 2021, 2024.
- The Lancet Digital Health.
- AISTATS 2021.
- IEEE Journal of Biomedical and Health Informatics.
- ISBI 2020.
- ICLR Workshop on Learning with Limited labeled Data (LLD 2019)
- MIDL 2019, 2020.
- Computer Vision and Image Understanding
- Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization
- NIPS Workshop on Learning with Limited labeled Data (LLD 2017)
- MICCAI Workshop on GRaphs in biomedicAl Image anaLysis (GRAIL 2017)
- XXI Congreso Argentino de Bioingeniería (SABI 2017)
- *10th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2016)*
- *IEEE Transactions on Medical Imaging*
- *Computerized Medical Imaging and Graphics*
- *Computer Vision and Image Understanding*
- *Journal of Imaging*

PROFESSIONAL MEMBERSHIPS

- MICCAI Society (2018)
- MICCAI Society (2014)
- Argentine Council of Engineering (Consejo Argentino de Ingenieros, CAI) (2013-2014)

OTHER MEETING ATTENDANCE

- MACI 2015 (5th Conference on Applied, Computational and Industrial Mathematics). UNICEN, Tandil, Argentina. May 4-6 2015.
- 2nd Biomedical Image Analysis Summer School: Modalities, Methodologies and Clinical Research, Institut Henri Pointcaré, Paris, France, July 8-12, 2013.
- ECImag 2011. 4th School and Workshop on Image Sciences, August 8-12, 2011.
- Agile Open Tandil 2011. Conferences on Agile methods for Software Development. Latin America Community of Agile Methodologies. Tandil, Argentina. May 21, 2011.
- MACI 2011 (3th Conference on Applied, Computational and Industrial Mathematics). UNS, Bahía Blanca, Argentina. May 9-11, 2011.

LECTURES, KEYNOTES & SCIENCE POPULARIZATION

- Curso de Formación Profesional “Fundamentos de Salud Digital & Inteligencia Artificial aplicados a la Transformación Digital en Telesalud”, Hospital Municipal “Dr. Leóndas Lucero”, Bahía Blanca, Argentina. April 2026.
- Invited lecturer, “Retinar: artificial intelligence and distributed telemedicine for early detection of diabetic retinopathy”.

Diplomatura en Economía de la Salud, Universidad de San Andrés, Argentina. November 7, 2025.

- Invited lecturer, “Retinar: artificial intelligence and distributed telemedicine for early detection of diabetic retinopathy”.

Cátedra Roche: Decisiones en Salud, Universidad de San Andrés, Argentina. September 17, 2025.

- Invited lecturer, “Retinar: artificial intelligence and distributed telemedicine for early detection of diabetic retinopathy”.

XXX Reunión de Comunicaciones Científicas y Tecnológicas 2025, UNNE, Resistencia, Argentina. Junio 2023

- Invited speaker for the inaugural lecture, entitled: “Artificial intelligence: from lab benches to the real world”.

Papers: Collage de Ciencia, Facultad de Arte, UNICEN, Tandil, Argentina. Noviembre 2023.

- Invited to participate in a [short video](#) popularizing scientific advancements from UNICEN.

DigitalHealth InvestorsTalk (podcast), Boston MA, USA. November 1st 2023.

- Invited to participate in Steven Wardell’s podcast to talk about “Checking in on AI in healthcare”.

Foro Salud Digital Bonaerense 2023. 18 de octubre de 2023.

- Keynote speaker, “retinar: teleophthalmology and AI”.

Papers: Collage de Ciencia, Facultad de Arte, UNICEN, Tandil, Argentina. October 9th 2023.

- Featured in a series of short videos popularizing scientific achievements from UNICEN.

Simposio Científico de Inteligencia Artificial y Aplicaciones (SCIAA), Universidad de San Andrés, CABA, Argentina. October 4th 2023.

- Keynote speaker, “One class to rule them all: applications of one class classification for computer vision and medicine.”

Del aula al ambito profesional, Centro de Estudiantes, Facultad de Ciencias Exactas, UNICEN, Tandil, Argentina. August 28th 2023.

- Member of the discussion panel, providing details about how to get your first job once you finish your university studies.

Congreso de Innovación en la Salud, FLAMA: Semana del Conocimiento, Tandil, Argentina. October 24th 2022.

- Keynote speaker, “retinar: telemedicine and artificial intelligence for preventing diabetic retinopathy.”

Seminario de Inteligencia Artificial en Medicina, Laboratorio de Inteligencia Artificial, Escuela de Negocios, Universidad Torcuato Di Tella, CABA, Argentina. September 21st 2022.

- Keynote speaker, “How to train a good baseline: experiences and recommendations from the application of deep learning in ophthalmology.”

Jornadas de Informática en Salud 2021 - Hospital Italiano de Buenos Aires, CABA, Argentina. November 16th 2021.

- Keynote speaker, “retinar: towards a national AI platform for early detection of diabetic retinopathy”.

	Reunión AIVO (Asociación de Investigación en Visión y Oftalmología) 2021 - Webinar. October 29th 2021. • Keynote speaker, “Artificial intelligence in ophthalmology: national developments, opportunities and challenges”. IMAGE AI, International Meeting of the European Glaucoma Society in glaucoma and artificial intelligence, Leuven, Belgium. December 12, 2019 • Lecturer, “What’s next in AI for glaucoma screening? The REFUGE challenge outcomes” II CONESEX, 2nd National Conference of Engineering Students, UNICEN, Tandil, Argentina. May 31, 2014 • Lecturer, “From math to medical image analysis: computational models applied in diagnosis, treatment and monitoring diseases” Shared workspace for students, Facultad de Ciencias Exactas, UNICEN, Tandil, Argentina. • Lecturer, “Introduction to Software Developments degrees”
APPLICATION AREAS	Fundus and OCT imaging, Computer-assisted diagnosis of ophthalmic diseases, Automated screening.
LANGUAGES	Spanish • Mother tongue. English • Fluent writing, reading and speaking. • Anglia Proficiency Level test approved with distinctions. French • Fluent reading, intermediate writing and speaking. • DELF A2 test with score 93/100. • 6 months spent in France. German • Basic reading, writing and speaking. • A1.1 level (abc Bildungszentrum, Vienna, Austria). • 1 year and 9 months spent in Austria.
TECHNICAL ENGINEERING AND AI SKILLS	• Tools: Visual Studio Code, Git, Matlab/Octave, Scilab, R, UML, Trello. • Programming Languages: Python, Matlab, Java, C++, SQL (mySQL, Oracle, PostgreSQL), Unix shell scripting, Markdown. • Computer Vision: Torchvision, OpenCV. • Machine learning: Pytorch, Tensorflow (keras), MatConvNet, Scikit Learn, Matlab, Pandas. • AI Research and planning: Identifying problems, gathering information from stakeholders (mostly clinicians), data recovery and curation, designing data annotation protocols, designing AI solutions for real world problems (mostly medical but also general computer vision/graphics too), establishing evaluation protocols, publishing results, consulting.
CITIZENSHIP	Argentina